Observer-Dependent Temporal Asymmetry: From Quantum Erasure to Gravitational Time

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The temporal asymmetry parameter $\tau = 1 - F$, defined via Petz recovery fidelity, depends on which degrees of freedom the observer can access. We formalize this by identifying each observer with a von Neumann subalgebra $\mathcal{A}_O \subseteq \mathcal{A}_{\text{total}}$, so that different observers assign different τ_O to the same physical process. We prove three results: (1) τ_O is monotone under observer refinement—an observer with access to more degrees of freedom always sees less temporal asymmetry (Theorem 1); (2) for any observer with a finite measurement budget $N_{\max}$, there exists an operational threshold $\epsilon_O = 1/(2N_{\max})$ below which the arrow of time is undetectable (Theorem 2); (3) combining two observers never increases temporal asymmetry beyond the minimum of the two (Theorem 3). We prove a complementary uncertainty relation: observers measuring mutually unbiased observables cannot simultaneously eliminate the arrow of time (Theorem 5). These results unify three domains under one framework: the quantum eraser (where different detector configurations yield $\tau = 0$ or $\tau > 0$ for the same entangled pair), black hole complementarity (where infalling and external observers assign different τ to the same process), and quantum error correction (where decoder quality determines the effective τ for logical information). Within this framework, the arrow of time is not an intrinsic property of the underlying dynamics, but rather reflects the observer's information access, as quantified by τ_O.

INTRODUCTION

The arrow of time is conventionally treated as a universal feature of nature: entropy increases, eggs do not unscramble, and the past is fundamentally different from the future. Yet the fundamental laws of quantum mechanics are unitary and time-symmetric. The standard resolution invokes decoherence [1, 2]: tracing over environmental degrees of freedom converts unitary evolution into an irreversible channel, producing entropy and breaking time-reversal symmetry.

This resolution already contains an observer-dependent element. Which degrees of freedom are "system" and which are "environment" is a choice. Different partitions—different observers—yield different amounts of irreversibility. The quantum eraser [3, 4] makes this vivid: an observer with access to only the signal photon sees decoherence and an arrow of time, while an observer with access to both signal and idler sees a pure entangled state and no arrow of time at all. The physics is identical; the temporal asymmetry is not.

In a companion paper [5], we defined the temporal asymmetry parameter $\tau = 1 - F(\rho, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ and showed that it connects quantum error correction, retrodiction, the quantum eraser, and thermodynamic entropy production through a four-way equivalence chain. Here we promote τ to an *observer-dependent* quantity τ_O and develop the consequences.

Each observer O is identified with a von Neumann subalgebra $\mathcal{A}_O$ of the total algebra of observables—the degrees of freedom O can access. The effective channel seen by O is the restriction to $\mathcal{A}_O$ (a partial trace over everything else), and the Petz recovery map constructed from this channel defines τ_O. The main results are:

1. *Monotonicity* (Theorem 1): More information access $\Rightarrow$ less temporal asymmetry. This follows from the data processing inequality applied to nested observer channels.

2. *Operational threshold* (Theorem 3): A finite-resource observer cannot detect the arrow of time below a threshold $\epsilon_O = 1/(2N_{\max})$, giving the arrow of time an operational, observer-dependent character.

3. *Subadditivity* (Theorem 4): Combining two observers always helps—or at worst does not hurt—for recovery.

4. *Complementary uncertainty* (Theorem 5): Observers measuring complementary observables cannot simultaneously eliminate the arrow of time, in analogy with the Maassen–Uffink entropic uncertainty relation [13].

The observer-dependent τ_O framework (Definitions 1–3 and Theorems 1–5) is the contribution of the present work. In particular, the complementary uncertainty relation (Theorem 5) and its tight bound $g(d) = (d+1) - \sqrt{2(d+1)}$ are new results, proved using known tools (the 2-design property of MUBs [34] and the Cauchy–Schwarz inequality) applied in a new way.

These results apply uniformly to the quantum eraser (Sec.), black hole complementarity (Sec.), and quantum error correction as observer hierarchy (Sec.). Section connects observer-dependent τ to the Wheeler–DeWitt equation, the Page–Wootters mechanism [9], and

the physical hierarchy of temporal asymmetry. We discuss connections to existing frameworks and open problems in Sec. .

FRAMEWORK

Definition 1 (Observer)**.** *An* observer O *is specified by a von Neumann subalgebra* $\mathcal{A}_O \subseteq \mathcal{A}_{\text{total}}$ *of the total algebra of observables. The subalgebra* $\mathcal{A}_O$ *represents the degrees of freedom accessible to* O*.*

Following Rovelli's relational quantum mechanics [10] and the quantum reference frame formalism of De Vuyst et al. [11], we use "observer" in a purely information-theoretic sense: any physical system that interacts with and records information about another system. No consciousness is required.

Definition 2 (Observer's channel)**.** *Given observer* O *with accessible algebra* $\mathcal{A}_O$*, the* observer's effective channel *is the conditional expectation*

$$\mathcal{N}_O : \mathcal{A}_{\text{total}} \to \mathcal{A}_O, \quad \mathcal{N}_O = \text{Tr}_{\mathcal{A}_O^c}, \tag{1}$$

obtained by tracing over all degrees of freedom not in $\mathcal{A}_O$*. When* $\mathcal{A}_O = \mathcal{B}(\mathcal{H}_S) \otimes \mathbf{1}_E$*, this reduces to the familiar partial trace* $\mathcal{N}_O = \text{Tr}_E$*.*

Definition 3 (Observer-dependent τ)**.** *The* temporal asymmetry as seen by observer O *is*

$$\tau_O = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}_O}(\mathcal{N}_O(\rho))\big), \tag{2}$$

where $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}_O}$ *is the (rotated) Petz recovery map [7, 35] for the channel* $\mathcal{N}_O$ *and reference state* σ*, and* F *is the Uhlmann fidelity.*

The JRSWW bound [7] immediately gives the observer-dependent entropy production bound:

$$\tau_O \leq 1 - \exp(-\Sigma_O/2), \tag{3}$$

where $\Sigma_O = D(\rho\|\sigma) - D(\mathcal{N}_O(\rho)\|\mathcal{N}_O(\sigma))$ is the relative entropy lost by restricting to observer O's accessible algebra.

Lattice of observers.—The set of observers forms a lattice under the refinement partial order $O_1 \leq O_2$ (defined by $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$):

- *Top* ($\top$): The total observer with $\mathcal{A}_\top = \mathcal{A}_{\text{total}}$ and $\mathcal{N}_\top = \text{id}$. Since no information is lost, $\tau_\top = 0$.
- *Bottom* ($\perp$): The trivial observer with $\mathcal{A}_\perp = \mathbb{C}\mathbf{1}$ (no information access). Here $\tau_\perp = 1 - F(\rho, \sigma)$ is the maximum possible temporal asymmetry.
- *Meet* ($O_1 \wedge O_2$): Access to $\mathcal{A}_{O_1} \cap \mathcal{A}_{O_2}$ (common information).
- *Join* ($O_1 \vee O_2$): Access to the algebra generated by $\mathcal{A}_{O_1} \cup \mathcal{A}_{O_2}$ (combined information).

MAIN THEOREMS

Theorem 1 (Monotonicity of τ under observer refinement)**.** *If* $O_1 \leq O_2$ *(i.e.,* $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$*), then*

$$\tau_{O_1}^{\text{opt}} \geq \tau_{O_2}^{\text{opt}}, \tag{4}$$

where $\tau_O^{\text{opt}} = 1 - \max_{\mathcal{R}} F(\rho, \mathcal{R} \circ \mathcal{N}_O(\rho))$ *is the temporal asymmetry under optimal recovery.*

Proof.—Since $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$, there exists a CPTP map (conditional expectation) $\mathcal{E}_{12} : \mathcal{A}_{O_2} \to \mathcal{A}_{O_1}$ such that $\mathcal{N}_{O_1} = \mathcal{E}_{12} \circ \mathcal{N}_{O_2}$.

For any recovery map $\mathcal{R}_1$ acting on $\mathcal{A}_{O_1}$, define $\mathcal{R}_1' = \mathcal{R}_1 \circ \mathcal{E}_{12}$ acting on $\mathcal{A}_{O_2}$. Then $F(\rho, \mathcal{R}_1 \circ \mathcal{N}_{O_1}(\rho)) = F(\rho, \mathcal{R}_1' \circ \mathcal{N}_{O_2}(\rho))$. Therefore

$$\begin{aligned} F_{\text{opt}}(\mathcal{N}_{O_1}) &= \max_{\mathcal{R}_1} F(\rho, \mathcal{R}_1 \circ \mathcal{N}_{O_1}(\rho)) \\ &\leq \max_{\mathcal{R}} F(\rho, \mathcal{R} \circ \mathcal{N}_{O_2}(\rho)) = F_{\text{opt}}(\mathcal{N}_{O_2}), \end{aligned} \tag{5}$$

since $\{\mathcal{R}_1 \circ \mathcal{E}_{12}\}$ is a subset of all recovery maps from $\mathcal{A}_{O_2}$. Hence $\tau_{O_1}^{\text{opt}} \geq \tau_{O_2}^{\text{opt}}$.

The same monotonicity holds for the JRSWW upper bound on τ: by the data processing inequality applied to $\mathcal{E}_{12}$, $\Sigma_{O_1} \geq \Sigma_{O_2}$, so $1 - \exp(-\Sigma_{O_1}/2) \geq 1 - \exp(-\Sigma_{O_2}/2)$. □

Corollary 2 (Entropy production is observer-dependent)**.** *For* $O_1 \leq O_2$*:* $\Sigma_{O_1} \geq \Sigma_{O_2}$*. A coarser observer always assigns more entropy production to the same physical process.*

Proof.—Immediate from the DPI applied to $\mathcal{E}_{12}$ in the proof above. □

This result is consistent with the independent finding of Basso, Maziero, and Celeri [12], who showed that entropy production for quantum systems in curved spacetime depends on the observer's worldline, and of De Vuyst et al. [11], who proved that gravitational entropy is observer-dependent via quantum reference frames.

Theorem 3 (Operational threshold)**.** *Let observer* O *have access to a POVM* $\{M_i\}_{i=1}^n$ *and a total measurement budget of* $N_{\max}$ *uses. If*

$$\tau_O < \epsilon_O \equiv \frac{1}{2N_{\max}}, \tag{6}$$

then no measurement available to O *can distinguish the forward process from its Petz reversal at a statistically significant level.*

Proof.—By the Fuchs–van de Graaf inequality [15], the total variation distance between the forward and retrodicted measurement distributions satisfies

$$\|p_{\text{fwd}} - p_{\text{ret}}\|_{\text{TV}} \leq \sqrt{1 - F^2} \leq \sqrt{2\tau_O}, \tag{7}$$

where the second inequality uses $1 - F^2 = \tau_O(2 - \tau_O) \leq 2\tau_O$.

By the Chernoff–Stein lemma, distinguishing two distributions at total variation distance δ with constant confidence requires $N \sim 1/\delta^2$ samples. Setting $\delta = \sqrt{2\tau_O}$ gives $N_{\text{needed}} \sim 1/(2\tau_O)$. When $\tau_O < 1/(2N_{\max})$, the required sample count exceeds $N_{\max}$, so the arrow of time is operationally undetectable. □

Remark.—Two observers with different measurement budgets can genuinely disagree on whether a given process has an arrow of time. The arrow of time is an operational concept, not merely a theoretical one.

Theorem 4 (Subadditivity under observer combination)**.** *For two observers O_1 and O_2 with joint observer $O_1 \vee O_2$:*

$$\tau^{\text{opt}}_{O_1 \vee O_2} \leq \min(\tau^{\text{opt}}_{O_1}, \tau^{\text{opt}}_{O_2}). \tag{8}$$

Proof.—Since $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_1 \vee O_2}$ and $\mathcal{A}_{O_2} \subseteq \mathcal{A}_{O_1 \vee O_2}$, Theorem 1 gives $\tau^{\text{opt}}_{O_1 \vee O_2} \leq \tau^{\text{opt}}_{O_1}$ and $\tau^{\text{opt}}_{O_1 \vee O_2} \leq \tau^{\text{opt}}_{O_2}$. □

Combining information from two observers always helps (or at worst does not hurt) for recovery. This is the Petz-recovery analogue of the general principle that pooling data improves inference.

Theorem 5 (Complementary observer uncertainty)**.** *Let $\{O_1, \ldots, O_{d+1}\}$ be observers whose accessible algebras correspond to a complete set of $d+1$ mutually unbiased bases (MUBs) in prime-power dimension d. For any pure state $|\psi\rangle$:*

$$\sum_{i=1}^{d+1} \tau_{O_i}(|\psi\rangle) \;\geq\; (d+1) - \sqrt{2(d+1)} \;=:\; g(d) \;>\; 0\,. \tag{9}$$

The bound is tight*: equality holds when the purity $P_i = \sum_k |\langle e_k^{(i)}|\psi\rangle|^4 = 2/(d+1)$ for all i. For qubits: $g(2) = 3 - \sqrt{6} \approx 0.551$.*

Proof.—Three ingredients. (i) For dephasing in basis i: $F_i = \sqrt{P_i}$, so $\tau_i = 1 - \sqrt{P_i}$. (ii) A complete set of MUBs forms a complex projective 2-design [34], giving $\sum_i P_i = 2$ for any $|\psi\rangle$. (iii) By Cauchy–Schwarz: $\sum_i \sqrt{P_i} \leq \sqrt{(d+1)\sum_i P_i} = \sqrt{2(d+1)}$. Therefore $\sum_i \tau_i = (d+1) - \sum_i \sqrt{P_i} \geq (d+1) - \sqrt{2(d+1)}$. Feasibility: $1/d \leq 2/(d+1) \leq 1$ holds for $d \geq 1$. Strict positivity: $g(d) = \sqrt{d+1}\,(\sqrt{d+1} - \sqrt{2}) > 0$ for $d \geq 2$. □

Physical interpretation.—This is the temporal analogue of the Maassen–Uffink entropic uncertainty relation [13]: just as total entropy across complementary measurements has a positive floor, *total irreversibility across complementary observers has a positive floor.* One cannot eliminate the arrow of time in all directions simultaneously. An observer who perfectly retrodicts position must be maximally ignorant about momentum, and vice versa. This is the Petz-recovery version of the Heisenberg uncertainty principle. Asymptotically, $g(d)/d \to 1$: in high dimension, the arrow of time is nearly maximal for almost every observer.

PHYSICAL APPLICATIONS

Quantum eraser

The quantum eraser [3, 4] is the cleanest illustration of observer-dependent τ. Consider the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where S is the signal (path) and E is the idler (which-path marker) with $\langle d_A|d_B\rangle = 0$.

Three observers see three different temporal asymmetries for the same physical process:

Observer 1 (signal only).—$\mathcal{A}_{O_1} = \mathcal{B}(\mathcal{H}_S) \otimes \mathbf{1}_E$. The channel is $\mathcal{N}_{O_1} = \text{Tr}_E$, giving the fully decohered state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. The Petz recovery map attempts to reconstruct $|\psi\rangle_{SE}$ from ρ_S alone. Since the entangled state is a purification and Tr_E maps it to a rank-2 mixture, recovery is imperfect: $F = 1/\sqrt{2}$, giving $\tau_{O_1} = 1 - 1/\sqrt{2} \approx 0.293$. Observer 1 sees an arrow of time.

Observer 2 (signal + idler).—$\mathcal{A}_{O_2} = \mathcal{B}(\mathcal{H}_S) \otimes \mathcal{B}(\mathcal{H}_E) = \mathcal{A}_{\text{total}}$. The channel is $\mathcal{N}_{O_2} = \text{id}$; the full state $|\psi\rangle_{SE}$ is directly accessible. Recovery is trivial: $F = 1$, $\tau_{O_2} = 0$. Observer 2 sees no arrow of time.

Observer 3 (idler only).—$\mathcal{A}_{O_3} = \mathbf{1}_S \otimes \mathcal{B}(\mathcal{H}_E)$. The channel is $\mathcal{N}_{O_3} = \text{Tr}_S$, giving $\rho_E = \frac{1}{2}(|d_A\rangle\langle d_A| + |d_B\rangle\langle d_B|)$. By the symmetry of the maximally entangled state, $\tau_{O_3} = 1 - 1/\sqrt{2} \approx 0.293$. Observer 3 also sees an arrow of time, equal to Observer 1's.

This pattern follows from Theorems 1 and 4: since $O_1 \leq O_2$ and $O_3 \leq O_2$, we have $\tau_{O_2} \leq \min(\tau_{O_1}, \tau_{O_3})$, which is satisfied with $0 \leq 0.293$. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of the Petz recovery map—it promotes Observer 1 to Observer 2 by granting access to the idler correlations.

The quantum eraser is not about "changing the past." It is about two observers assigning different τ to the same physics because they access different subalgebras. The interference pattern was always present in $|\psi\rangle_{SE}$; only Observer 1's restricted access made it invisible.

Black hole complementarity

Black hole complementarity [17] states that infalling and external observers give complementary but individually consistent descriptions of the same physics. In the observer-dependent τ framework, this becomes quantitative.

Infalling observer.—An observer who freely falls through the horizon accesses the local algebra near the horizon, which includes both interior and exterior modes

via the Unruh effect. For an observer crossing the horizon of a Schwarzschild black hole, the local evolution is approximately unitary (by the equivalence principle), so $\tau_{\text{in}} \approx 0$.

External observer.—An observer who remains at fixed radius r outside the horizon traces over the interior modes. The effective channel is a thermal channel at the Hawking temperature $T_H = 1/(8\pi GM)$. The entropy production is $\Sigma_{\text{ext}} = -\ln(-g_{00}) \approx r_s/r$ for Schwarzschild [6], giving

$$\tau_{\text{ext}} \leq 1 - \exp(-r_s/(2r)). \tag{10}$$

As $r \to r_s$, $\tau_{\text{ext}} \to 1$: the external observer sees maximal temporal asymmetry for processes near the horizon. As $r \to \infty$, $\tau_{\text{ext}} \to 0$: far from the black hole, spacetime is flat and the arrow of time from gravity vanishes.

Same physical process. Different observer algebras. Different τ. This is the same structure as the quantum eraser, with the horizon playing the role of the entanglement boundary.

The recent work of Basso, Maziero, and Celeri [12] makes this precise: they derive an observer-dependent quantum fluctuation theorem in curved spacetime, showing that the entropy production (and hence τ) depends on the observer's worldline through the local Hamiltonian $H(\tau) = H_0 + ma_i x^i + \frac{m}{2} R_{\tau i \tau j} x^i x^j$.

QEC as observer hierarchy

In quantum error correction, the decoder is an observer. The noise channel $\mathcal{N}$ degrades the encoded state; the decoder's recovery map $\mathcal{R}_{\text{dec}}$ attempts to reconstruct it. The temporal asymmetry for the decoded logical information is $\tau_{\text{dec}} = 1 - F(\rho_L, \mathcal{R}_{\text{dec}} \circ \mathcal{N}(\rho_L))$—the logical error rate.

Different decoders, accessing the same syndrome data but processing it differently, achieve different τ:

$$\tau_{\text{ML}} \leq \tau_{\text{NN}} \leq \tau_{\text{MWPM}} \leq \tau_{\text{UF}}, \tag{11}$$

where ML is maximum likelihood, NN is the neural-network decoder of Ref. [21], MWPM is minimum-weight perfect matching, and UF is Union-Find [20]. This hierarchy [5] is consistent with each decoder approximating the Petz map to a different degree, since the Petz map is the unique Bayesian-consistent retrodiction [8].

A perfect decoder achieves $\tau_{\text{dec}} = 0$ for the protected subspace—it eliminates the arrow of time for the logical qubit even though the physical qubits interact with a noisy environment. This is exactly Theorem 1 in action: the decoder enlarges the observer's effective algebra (from raw noisy qubits to error-corrected logical qubits), reducing τ.

In the holographic context [18, 19], the entanglement wedge $W(A)$ of a boundary region A is precisely the set of bulk operators for which $\tau = 0$ when the boundary observer has access to A. Operators outside $W(A)$ have $\tau > 0$. The Ryu–Takayanagi surface marks a sharp phase transition: as the boundary region grows past the critical size $|A| = |\partial|/2$, deep bulk operators jump from $\tau = 1$ to $\tau = 0$.

TABLE I. Observer hierarchy of temporal asymmetry. Each level traces out more degrees of freedom, increasing τ.

Observer	τ	**Accessible algebra**
Universe (WdW)	0	$\mathcal{A}_{\text{total}}$
QC (error-corrected)	$\epsilon \to 0^+$	System + ancilla
Fine-grained detector	$\tau_{\text{fine}} \ll 1$	System + partial env.
Coarse-grained observer	$\tau_{\text{coarse}} \sim O(1)$	Classical degrees of freedom
Thermodynamic limit	$1 - e^{-\Sigma/2}$	Macroscopic averages

THE ARROW OF TIME AS OBSERVER ARTIFACT

The Wheeler–DeWitt equation $H|\Psi\rangle = 0$ for a closed universe implies $\tau_{\text{universe}} = 0$: the total state evolves unitarily (trivially—it is an eigenstate of H), so there is no arrow of time at the fundamental level [22]. Time emerges through the Page–Wootters mechanism [9]: an observer uses a subsystem as a clock and conditions the remaining state on the clock reading, producing an effective time-dependent state with $\tau > 0$.

De Vuyst et al. [11] recently proved that the Page–Wootters formalism is mathematically equivalent to the quantum reference frame (QRF) approach. Different clock choices (different QRFs) give different conditional states and different τ_O. This is observer-dependent temporal asymmetry at the cosmological level.

The physical hierarchy of τ values is (Table I):

This hierarchy follows as a direct consequence of Theorem 1. Each step down the hierarchy corresponds to tracing out more degrees of freedom, which by the DPI can only increase Σ_O and hence increase the upper bound on τ_O. The key relationships are:

$\tau_{\text{universe}} = 0$.—For the total observer of a closed universe, the global evolution is unitary. No information is lost, no entropy is produced. The arrow of time does not exist at the fundamental level.

$\tau_{\text{subsystem}} > 0$.—Any subsystem observer traces out part of the universe. By Theorem 1, $\tau > 0$ for generic non-product states. This is Zurek's einselection [1]: environment-induced decoherence produces entropy ($\Sigma > 0$), forcing $\tau > 0$. In our language, *decoherence is the observer's τ becoming positive.*

$\tau_{\text{fine}} \leq \tau_{\text{coarse}}$.—A fine-grained detector monitoring environmental degrees of freedom has access to a larger subalgebra than a coarse-grained observer restricted to

macroscopic variables. By Theorem 1, $\tau_{\text{fine}} \leq \tau_{\text{coarse}}$. The gap can be large when the environment contains recoverable quantum correlations that the fine-grained detector can resolve but the coarse-grained observer cannot.

In this framework, the arrow of time may be understood as the cost of incomplete information access, quantified by τ_O.

DISCUSSION

Connection to existing frameworks.—The observer-dependent τ framework is consistent with several existing approaches, which it makes quantitative. Rovelli's relational QM [10] holds that physical quantities are relational—defined only with respect to a reference system. Our τ_O is the quantitative version: different observers (reference systems) assign different temporal asymmetries to the same process. Zurek's einselection [1] explains why classical reality appears: environmental monitoring selects a pointer basis. In τ language, a state is "classical" when τ_O converges across all observers who sample any environmental fragment—this is quantum Darwinism expressed as τ convergence.

Friston's free energy principle [23] states that persistent systems minimize variational free energy. Since the variational free energy $\mathcal{F}$ is an upper bound on surprise $-\ln p(y)$, and surprise corresponds to the irrecoverable entropy production Σ, free energy minimization is structurally related to τ-minimization. The Petz recovery map is the quantum Bayes rule [8, 16], so optimal Bayesian inference is Petz recovery. The quantitative bridge between Friston's $\mathcal{F}$ (a KL divergence) and τ (a fidelity measure) remains to be established rigorously; this is one of the open problems.

Testable predictions.—The framework yields several concrete, quantitative predictions.

(1) *Observer-dependent τ for the same process.* Different measurement apparatuses should yield different τ for the same physical process. This is already demonstrated qualitatively in the quantum eraser, but the prediction is quantitative: for a maximally entangled qubit pair ($d = 2$), Observer 1 (signal only) should obtain $\tau_{O_1} = 1 - 1/\sqrt{2} \approx 0.293$, while Observer 2 (signal + idler) should obtain $\tau_{O_2} = 0$. In a superconducting qubit experiment, an observer who monitors the cavity environment (via a readout resonator) should measure a smaller τ than an observer who monitors only the qubit register.

(2) *Complementary uncertainty for qubit MUBs.* For a single qubit ($d = 2$), the three MUBs are the eigenbases of σ_x, σ_y, and σ_z. For any pure qubit state $|\psi\rangle$, the sum of dephasing-induced temporal asymmetries satisfies $\sum_{i=1}^{3} \tau_{O_i}(|\psi\rangle) \geq 3 - \sqrt{6} \approx 0.551$. The bound is saturated when $P_i = 2/3$ for all three bases, corresponding to states on the circle $|\langle e_k^{(i)}|\psi\rangle|^2 = (1 \pm 1/\sqrt{3})/2$. This can be tested on ion-trap [25] or NMR [26] platforms by preparing a known pure state, applying dephasing in each of the three Pauli bases, and measuring the recovery fidelity.

(3) *Operational threshold.* For a process with $\tau_O = 0.01$, Theorem 3 predicts that at least $N_{\text{needed}} \sim 50$ measurement repetitions are required to distinguish the forward process from its Petz reversal. A process with $\tau_O = 10^{-4}$ requires $N \sim 5000$. These sample counts can be compared against experimental data in existing Petz-map implementations [25, 26].

(4) *Monotonicity under partial environmental access.* Consider a qubit undergoing amplitude damping with damping parameter γ. An observer with access to the qubit alone sees $\tau_{O_1} = 1 - \sqrt{1-\gamma}$. An observer with additional access to the emitted photon (environment) should see $\tau_{O_2} < \tau_{O_1}$. For $\gamma = 0.5$, $\tau_{O_1} \approx 0.293$, while $\tau_{O_2} = 0$ when the full purification is accessible. Intermediate environmental access should interpolate monotonically between these values, as predicted by Theorem 1.

Scope and limitations.—Theorem 1 is proved for optimal recovery; the statement for the standard Petz map itself (rather than the optimum over all recovery maps) is a conjecture supported by all known examples. The complementary uncertainty relation (Theorem 5) is proved for *pure states* in prime-power dimension d (where complete MUB sets exist) via the 2-design purity sum rule and Cauchy–Schwarz; the exact bound $g(d) = (d+1) - \sqrt{2(d+1)}$ is tight and stronger than an earlier conjectured value of $g(2) = 1/2$ (the exact value is $3 - \sqrt{6} \approx 0.551$). Two extensions remain open: (a) the generalization of Theorem 5 to *mixed states*, where the purity sum rule requires modification; and (b) the extension to *non-prime-power dimensions*, where complete MUB sets are not known to exist and the 2-design argument does not directly apply. The black hole application (Sec.) uses the gravitational entropy production $\Sigma_{\text{grav}} = -\ln(-g_{00})$ derived in Ref. [6] via the gravitational thermal attenuator channel; readers should note that this gravitational result itself awaits independent verification and experimental test.

Open questions.—(i) Extend Theorem 5 to mixed states and to non-prime-power dimensions where complete MUB sets are not known to exist. (ii) Extend the framework to non-Markovian dynamics [24], where Σ_O can be temporarily negative, implying transient decreases in τ_O. (iii) Compute τ_O explicitly for a Page–Wootters toy model with different clock choices, connecting to De Vuyst et al. [11]. (iv) Quantify the relationship between Friston's free energy [23] and τ_O. (v) Extend Theorem 1 from optimal recovery to the Petz map itself.

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